where $c_{11} = [0 \;\; c \;\; b]\begin{bmatrix}0\\c\\b\end{bmatrix} = 0\cdot 0 + c\cdot c + b\cdot b = c^2 + b^2$ and other elements c_{ij} may be computed similarly.

Finally, we easily obtain that

$$A^2 = \begin{bmatrix} 0+c^2+b^2 & 0+0+ab & 0+ac+0 \\ 0+0+ab & c^2+0+a^2 & bc+0+0 \\ 0+ac+0 & bc+0+0 & b^2+a^2+0 \end{bmatrix} = \begin{bmatrix} b^2+c^2 & ab & ac \\ ab & c^2+a^2 & bc \\ ac & bc & a^2+b^2 \end{bmatrix}.$$

Example 7.9

Solve for x if $[x \quad 2 \quad -1]\begin{bmatrix}1 & 1 & 2\\ -1 & -4 & 1\\ -1 & -1 & -2\end{bmatrix}\begin{bmatrix}x\\2\\1\end{bmatrix} = O.$

Solution

$$[x \quad 2 \quad -1]\begin{bmatrix}1 & 1 & 2\\ -1 & -4 & 1\\ -1 & -1 & -2\end{bmatrix}\begin{bmatrix}x\\2\\1\end{bmatrix} = O$$

That is, $$[x-2+1 \quad x-8+1 \quad 2x+2+2]\begin{bmatrix}x\\2\\1\end{bmatrix} = O$$

$$[x-1 \quad x-7 \quad 2x+4]\begin{bmatrix}x\\2\\1\end{bmatrix} = O$$

$$x(x-1)+2(x-7)+1(2x+4) = 0$$

$$x^2+3x-10 = 0 \Rightarrow x = -5, 2.$$

Note 7.3

We have the following important observations:

(1) If $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{n\times p}$, and $m \neq p$, then the product AB is defined but not BA.

(2) The fundamental properties of real numbers namely,

$ab = ba \quad \forall a, b \in \mathbb{R}$

$ab = ac \Rightarrow b = c \;\forall\; a, b, c \in \mathbb{R},\; a \neq 0$

$ab = 0 \Rightarrow a = 0$ or $b = 0 \;\forall\; a, b \in \mathbb{R}.$

Can we discuss these in matrices also?

(i) Even if AB and BA are defined, then $AB = BA$ is not necessarily true.

For instance, we consider

$$A = \begin{bmatrix}1 & 1\\ 2 & 0\end{bmatrix} \text{ and } B = \begin{bmatrix}2 & -1\\ 3 & 1\end{bmatrix}$$

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